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Sets

A set is a well-defined collection of objects.

Representation of Set :-

(i) Roster form :-

Set of natural number

$A = \{1, 2, 3, 4, \ldots\}$

(ii) Set builder form :-

$A = \{x : x \text{ is a natural number}\}$

* Types of set

(i) Null or empty set (ϕ) :- That contains no element.

(ii) Finite or Infinite set :- If the elements are countable, then finite otherwise it is infinite set.

* Singleton sets :- A set containing exactly 1 element is called singleton set.

For ex :- $\{x : x \in N \ \& \ x^2 - 9 = 0\} = \{3\}$

* Subset :-

A & B

If $A = \{1, 2\}$

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$B = \{1, 2, 3, 4\}$
$A \subset B$ (A is a subset of B)
Elements of A is contained in B. Therefore A is a subset of B & B is a superset of A because B contains elements of its superior.

Types of Subset
(i) Proper subset (when A & B are not equal)
(ii) Improper subset
Ex. of Improper subset
$A = \{1, 2\}$
$B = \{1, 2\}$
$A \subseteq B$ or $B \subseteq A$
Improper subset exists only when two sets (A or B) are equal.

* Power set :- The set of all subsets of a given set(A) is called power set of A represented by $P(A)$
For ex:- $A = \{1, 2\}$
$P(A) = \{\emptyset, \{1\}, \{2\}, \{1, 2\}\}$
Number of elements in power set :- 2^n the elements of sets given.

* Operation of Sets :-

(i) Union

$A \cup B = \{x : x \in A \text{ or } x \in B\}$
Ex:- $A = \{1, 2, 3\}$
$B = \{3, 4, 5, 6\}$
$A \cup B = \{1, 2, 3, 4, 5, 6\}$

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(ii) Intersection:-

$A \cap B = \{x : x \in A \ \& \ x \in B\}$

Ex:- $A = \{1, 2, 3\}$

$B = \{3, 4, 5, 6\}$

$A \cap B = \{3\}$

(iii) Difference of sets:-

$A - B = \{x : x \in A \ \& \ x \notin B\}$

$B - A = \{x : x \in B \ \& \ x \notin A\}$

A-B :- Ex:- $A = \{1, 2, 3\}$

$B \{3, 4, 5, 6\}$

$A - B = \{1, 2\}$

B-A :- $\{4, 5, 6\}$

(iv) Complement of a set :- Let U is a universal set and A is a subset of U, then the complement of A denoted by A' or A^c is given by

$A' = \{x : x \in U \ \& \ x \notin A\}$

Let $U = \{1, 2, 3, 4, 5, 6, 7, 8\}$

$A = \{1, 2, 3\}$

$A' = \{4, 5, 6, 7, 8\}$

* Algebra of Set theory:-

(i) Commutativity:-

$A \cup B = B \cup A$

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$A \cap B = B \cap A$

(ii) Associativity:-

Ex:- $1+(2+3) = (1+2)+3$

In terms of set $A \cup (B \cup C) = (A \cup B) \cup C$

$A \cap (B \cap C) = (A \cap B) \cap C$

(iii) Distributivity

(a) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ (Union distributes over intersection)

(b) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ (intersection distributes over union)

* Demorgan's law

(a) $(A \cup B)' = A' \cap B'$

(b) $(A \cap B)' = A' \cup B'$

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Proof for Demorgan's Law

$(A \cup B)' \subset A' \cap B'$

Let x belongs to $(A \cup B)'$ i.e. $x \in (A \cup B)'$
that means $x \notin A \cup B$

$\Rightarrow x \notin A \cup B$
$\Rightarrow x \notin A$ & $x \notin B$
$\Rightarrow x \in A'$ & $x \in B'$
$\Rightarrow x \in A' \cap B'$

$\because x \in A \cup B$
$\Rightarrow x \in A$ or $x \in B$
$x \notin A$ & $x \notin B$

Conversely, we prove that
Let $y \in A' \cap B'$
$\Rightarrow y \in A'$ & $y \in B'$
$\Rightarrow y \notin A$ & $y \notin B$
$\Rightarrow y \notin A \cup B$
$\Rightarrow y \in (A \cup B)'$

Therefore, $(A \cup B)' \subset A' \cap B'$

Qn: Prove that $A - (B \cup C) = (A - B) \cap (A - C)$
Soln: Let $x \in A - (B \cup C)$
$\Rightarrow x \in A$ & $x \notin (B \cup C)$ By the definition of difference of sets

$\Rightarrow x \in A$ & $(x \notin B$ & $x \notin C)$
$\Rightarrow (x \in A$ & $x \notin B)$ & $(x \in A$ & $x \notin C)$
$\Rightarrow x \in A - B$ & $x \in A - C$
$\Rightarrow x \in (A - B) \cap (A - C)$
$\therefore A - (B \cup C)$ is a subset of $(A - B) \cap (A - C)$

Conversely, we prove that

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Let $y \in (A-B) \cap (A-C)$

$\Rightarrow y \in A-B$ & $y \in A-C$

$\Rightarrow (y \in A$ & $y \notin B)$ & $(y \in A$ & $y \notin C)$

$\Rightarrow y \in A$ & $(y \notin B$ & $y \notin C)$

$\Rightarrow y \in A$ & $y \notin B \cup C$

$\Rightarrow y \in A - (B \cup C)$

$\therefore A - (B \cup C) = (A-B) \cap (A-C)$

Qn2: $A - (B \cap C) = (A-B) \cup (A-C)$

Qn3: $A \cap (B-C) = (A \cap B) - (B \cap C)$

Soln: Let $x \in A \cap (B-C)$

$\Rightarrow x \in A$ & $x \in (B-C)$

$\Rightarrow x \in A$ & $(x \in B$ & $x \notin C)$

$\Rightarrow (x \in A$ & $x \in B)$ & $(x \in B$ & $x \notin C)$

$\Rightarrow x \in A \cap B$ & $x \notin B \cap C$

$\Rightarrow x \in (A \cap B) - (B \cap C)$

Conversely, we prove that, we get

Let $y \in (A-B) \cup (A-C)$

Let $y \in (A \cap B) - (B \cap C)$

$\Rightarrow y \in (A \cap B)$ & $y \notin (B \cap C)$

$\Rightarrow (y \in A$ & $y \in B)$ & $(y \notin B$ or $y \notin C)$

$\Rightarrow y \in A$ & $(y \in B$ & $y \notin C)$

$\Rightarrow y \in A$ & $y \in (B-C)$

$\Rightarrow y \in A \cap (B-C)$

Qn4: $A \cup (B-C) = (A \cup B) - (A \cup C)$

Qn5: Prove that $(A-B) = A \cap B'$

Soln: Let $x \in A-B$

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$\Rightarrow x \in A \ \& \ x \notin B$

$\Rightarrow x \in A \ \& \ x \in B'$

$\Rightarrow x \in A \cap B'$

Now let $y \in A \cap B'$

$\Rightarrow y \in A \ \& \ y \in B'$

$\Rightarrow y \in A \ \& \ y \notin B$

$\Rightarrow y \in A - B$

H.W. Qn2 = $A-(B \cap C) = (A-B) \cup (A-C)$

Let $x \in A-(B \cap C)$

$x \in A \ \& \ x \notin (B \cap C)$

$\Rightarrow x \in A \ \& \ x \notin B$ or $x \notin C$

$\Rightarrow (x \in A \ \& \ x \notin B)$ or $(x \in A \ \& \ x \notin C)$

$\Rightarrow (x \in A - B)$ or $(x \in A - C)$

$\Rightarrow (A-B) \cup (A-C)$

Let $y \in (A-B) \cup (A-C)$

$(y \in A \ \& \ y \notin B)$ or $(y \in A \ \& \ y \notin C)$

$y \in A \ \& \ (y \notin B$ or $y \notin C)$

$y \in A \ \& \ y \notin (B \cap C)$

$y \in A - (B \cap C)$

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Unit-1

Qnt= For any two sets A and B. Prove that
$A \cap (B - A) = \phi$

Soln=

$A \cap (B-A) = \{x : x \in A \ \& \ x \in (B-A)\}$
$= \{x : x \in A \ \& \ x \in B \ \& \ x \notin A\} = \phi$
$\because$ ~~There~~ There will be no such element such that $x \in A \ \& \ x \notin A$

* Cartesian Product Of two Sets

If A and B are any two sets, then the set of odd ordered pairs of the elements of A&B is called the cartesian product of sets A&B. denoted by $A \times B$ and is given by
$A \times B = \{(a, b) : a \in A \ \& \ b \in B\}$

Ex:- if $A = \{1, 2, 3\}$ & $B = \{x, y\}$
$A \times B = \{(1, x) (1, y), (2, x) (2, y) (3, x) (3, y)\}$

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